

Basics 7: Fourier Analysis and Normal Modes

I - Mathematical and computational foundations

Spatial harmonics, wave numbers, dispersion relations, toroidal mode labels, and coupling

Textbook draft, version 1.0. Fourier analysis expresses fields as sums of simple oscillations. In toroidal plasmas, poloidal and toroidal harmonics reveal how geometry couples wave components. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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12 References**5****1 Learning objectives**

After completing this note, the reader should be able to:

- Derive Fourier-series coefficients from orthogonality.
- Interpret Fourier transforms and wave number.
- Use complex exponentials to represent real fields.
- Derive normal-mode dispersion relations.
- Interpret poloidal and toroidal mode numbers.
- Recognize aliasing, spectral resolution, and harmonic coupling.

2 Prerequisites and reading strategy

- Basics 2, 5, and 6.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Fourier series and orthogonality

For a 2π -periodic field,

$$f(\theta) = \sum_{m=-\infty}^{\infty} \hat{f}_m e^{im\theta}. \quad (3.1)$$

Orthogonality gives

$$\hat{f}_m = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-im\theta} d\theta. \quad (3.2)$$

A real field obeys $\hat{f}_{-m} = \hat{f}_m^*$. Parseval relation connects physical-space and spectral energy:

$$\frac{1}{2\pi} \int_0^{2\pi} |f|^2 d\theta = \sum_m |\hat{f}_m|^2. \quad (3.3)$$

4 Fourier transforms and differentiation

For a nonperiodic coordinate,

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad (4.1)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk. \quad (4.2)$$

Derivatives become multiplication:

$$\widehat{f_x} = ik\hat{f}, \quad \widehat{f_{xx}} = -k^2\hat{f}. \quad (4.3)$$

This converts constant-coefficient linear PDEs into algebraic equations for each wave number.

5 Normal modes and dispersion

Assume

$$u(\mathbf{x}, t) = \tilde{u} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}. \quad (5.1)$$

Then $\partial_t \rightarrow -i\omega$ and $\nabla \rightarrow i\mathbf{k}$. A nonzero amplitude requires a dispersion relation

$$D(\omega, \mathbf{k}) = 0. \quad (5.2)$$

For the wave equation, $\omega^2 = c^2 k^2$. Phase velocity is ω/k and group velocity is $d\omega/dk$.

6 Poloidal and toroidal harmonics

On a torus,

$$f(s, \theta, \phi, t) = \sum_{m,n} \hat{f}_{mn}(s) e^{i(m\theta - n\phi - \omega t)}. \quad (6.1)$$

m is the poloidal mode number and n toroidal under this convention. Other sign conventions exist. The radial amplitude is not generally sinusoidal because equilibrium quantities and boundaries vary with s .

7 Geometric harmonic coupling

In a uniform medium, Fourier harmonics can evolve independently. A coefficient

$$C(\theta) = C_0 + \epsilon \cos \theta \quad (7.1)$$

multiplied by $e^{im\theta}$ produces $m - 1$, m , and $m + 1$ components. Toroidicity therefore couples neighboring poloidal harmonics. Three-dimensional stellarator geometry contains many equilibrium harmonics and generates broader coupling networks.

8 Sampling, aliasing, and spectral interpretation

With N samples, only modes below the Nyquist limit are represented uniquely. Higher modes appear as aliases. Nonlinear products generate harmonics above the original bandwidth, motivating dealiasing in spectral solvers.

A physical eigenmode can contain many Fourier harmonics. The labels (m, n) describe components, not necessarily distinct independent modes.

Always record the sign convention and array ordering for Fourier coefficients. A silent convention mismatch can reverse $k_{\parallel}$ and corrupt resonance analysis.

9 Computational laboratory

Sample a signal containing $m = 3$ and $m = 7$, compute its discrete Fourier transform, and recover amplitudes and phases. Reduce the grid to demonstrate aliasing. Multiply a pure harmonic by $1 + \epsilon \cos \theta$ and verify generation of neighboring harmonics. Compare Python and MATLAB FFT ordering.

10 Summary

- Fourier analysis decomposes periodic fields into orthogonal harmonics.
- Normal-mode substitution reduces linear PDEs to dispersion or eigenvalue problems.
- Toroidal fields use poloidal and toroidal mode numbers.
- Geometry couples harmonics through nonuniform coefficients.
- Sampling and conventions determine whether spectra are interpreted correctly.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
m, n	mode numbers	Poloidal and toroidal indices.
$\hat{f}_m$	Fourier coefficient	Complex amplitude of harmonic m .
$\mathbf{k}$	wave vector	Spatial phase gradient.
ω	angular frequency	Temporal phase rate.
v_p, v_g	velocities	Phase and group velocity.

11 Exercises

1. Derive the Fourier coefficient formula from orthogonality.
2. Prove the derivative property of the Fourier transform.
3. Derive phase and group velocity for $\omega^2 = \omega_0^2 + c^2 k^2$.
4. Interpret constant-phase surfaces for an (m, n) toroidal harmonic.
5. Explain the conjugate symmetry of a real field.

12 References

References

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